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# STATIS and Dynamic Biplot Approaches for Socio-Economic Spatio-Temporal Analysis in Western Java 2020–2024

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## Abstract (TNR, 14pt)

The strategic regions of Banten, DKI Jakarta, and West Java, which form the core of Indonesia's national economy, experienced complex structural changes during 2020–2024 as a consequence of the COVID-19 pandemic and an uneven recovery across regencies and cities. Conventional multivariate approaches are limited in capturing the simultaneous interaction between spatial units, socio-economic indicators, and temporal shifts within a unified framework. This study applies the STATIS (Structuration des Tableaux à Trois Indices de la Statistique) method, integrated with Dynamic Biplot visualization, to evaluate longitudinal stability and map the spatio-temporal trajectories of eleven strategic development indicators across 41 regencies/cities. The analysis stages include demographically weighted Z-score standardization, inter-structure evaluation using RV coefficients, construction of a compromise structure through optimal spectral weighting, Monte Carlo permutation testing, and K-Means clustering (k = 3) on the compromise coordinates. The results show very high inter-year consistency (all RV coefficients above 0.95) and a statistically significant compromise structure ($p = 0.001$), whose first two dimensions explain 71.88% of total variance and represent the level of human and economic development and labor market pressure, respectively. The Dynamic Biplot identifies three trajectory patterns: V-shaped recovery in service-based metropolitan centers, stagnation in the southern regencies of Banten and West Java, and progressive growth in buffer cities. The three resulting clusters (Underdeveloped, Developing, Advanced) align strongly with UNDP HDI categories, indicating that regional disparities are driven by persistent structural factors and require targeted policy interventions.

**Keywords**: Interactive Visualization, K-Means Clustering, Longitudinal Data, PCA, Three-Way Data Analysis.

## 1. Introduction

The strategic region comprising Banten, DKI Jakarta, and West Java Provinces serves not only as the engine of Indonesia's national economic growth, contributing an average share of approximately 33.8 percent of the national GDP (Badan Pusat Statistik, 2025b), but also as an arena of highly complex human development dynamics over the past five years. Development success in this region can no longer be measured solely through macroeconomic indicators; it involves multidimensional interactions among quality of life as reflected in the Human Development Index (HDI), the Gender Development Index (GDI), and social welfare challenges such as poverty and food security that continue to evolve. This complexity demands deeper empirical evidence on how these indicators are actually distributed across regencies and municipalities.

Empirically, the development landscape of the three provinces exhibits sharp polarization between principal growth centers and peripheral areas. Although the GRDP of major cities records strong economic performance, such achievement has not been accompanied by an even distribution of welfare. This is evident from the striking spatial variation in poverty, where the Percentage of Poor Population (PPM) ranges widely from 2.29 percent to 15.06 percent, and from differences in the Poverty Line (GK) and the Prevalence of Undernourishment (PKKP) that vary considerably across regions (Badan Pusat Statistik, 2024). Human capability gaps are also pronounced: Mean Years of Schooling (RLS) and Life Expectancy (AHH) still differ markedly across regions, while the Open Unemployment Rate (TPT) fluctuated between 6.55 and 10.12 percent during 2020–2024.

Beyond spatial disparities, the analytical challenge is amplified by highly volatile temporal dynamics throughout 2020–2024. The period began with the COVID-19 pandemic, which simultaneously depressed macroeconomic, labor-market, and welfare indicators (Biegert et al., 2023) — economic growth contracted to −2.1 percent in 2020 (Badan Pusat Statistik, 2025b) — followed by a recovery whose pace was uneven across regions. This drastic shift from crisis to recovery is strongly suspected to have altered the structural relationships among indicators; consequently, classical multivariate approaches that assume a static relationship structure over time are no longer adequate for analyzing this fluctuating period.

To overcome this limitation, a three-way data analysis approach is required, one capable of handling complex data structures consisting of objects (regencies/municipalities), variables (socio-economic indicators), and occasions (time). This study employs the STATIS method (Structuration des Tableaux à Trois Indices de la Statistique), an exploratory technique introduced by Lavit et al. (1994) and further reviewed by Oliveira et al. (2024), which constructs a compromise structure representing the common pattern of inter-indicator relationships across the observation period. The STATIS analysis is integrated with Dynamic Biplot visualization (Gabriel, 1971; Egido & Galindo, 2015), which projects regions and indicators into a low-dimensional space while tracing the trajectory of positional shifts from year to year, thereby distinguishing short-term pandemic fluctuations from long-term structural change.

This exploratory multivariate application fills an important gap in the Indonesian regional analysis literature, which has been dominated by classical inferential approaches rooted

in panel-data econometrics (Baltagi, 2021), such as dynamic spatial Durbin panel models for poverty (Nuswantari, 2021) and spatial epidemiological modeling (Ahmad, 2024) that focus on estimating the effects of explanatory variables rather than mapping the simultaneous inter-variable structure and its evolution. Accordingly, this study aims to: (1) characterize the compromise structure representing the multivariate relationships of eleven development indicators across 41 regencies/municipalities; (2) statistically test the longitudinal stability of the inter-indicator structure during the pandemic (2020–2021) and post-pandemic (2022–2024) periods using a permutation test; (3) map the trajectories of regional socio-economic change through Dynamic Biplot visualization; (4) identify annual spatial clustering patterns and inter-cluster migration dynamics; and (5) develop a web-based interactive exploration tool using the Dash and Plotly frameworks.

## 2. Research Method

This research adopts a quantitative approach with a descriptive-exploratory design, utilizing advanced multivariate (three-way) analysis to explore the longitudinal data structure of strategic development indicators. The data are secondary data collected through documentation of official Badan Pusat Statistik (BPS) publications, primarily the annual "Province in Figures" series for Banten, DKI Jakarta, and West Java for 2020–2024 (Badan Pusat Statistik Provinsi Banten, 2021, 2022, 2023, 2024; Badan Pusat Statistik Provinsi DKI Jakarta, 2021, 2022, 2023, 2024; Badan Pusat Statistik Provinsi Jawa Barat, 2021, 2022, 2023, 2024). The observation units cover all 41 regencies/municipalities in the three provinces (complete enumeration; 8 in Banten, 6 in DKI Jakarta, and 27 in West Java) across K = 5 annual occasions.

Eleven welfare indicators serve as analysis inputs: the Gender Development Index (IPG), Percentage of Poor Population (PPM), Gross Regional Domestic Product at Constant Prices (PDRB ADHK), Mean Years of Schooling (RLS), Male Life Expectancy (AHH L), Female Life Expectancy (AHH P), Poverty Line (GK), Prevalence of Undernourishment (PKKP), Open Unemployment Rate (TPT), Labor Force Participation Rate (TPAK), and Gini Ratio (GR). Two auxiliary variables are collected but excluded from the STATIS input: the Human Development Index (IPM), reserved as an external benchmark against the normative UNDP classification, and Population (JP), used to construct the demographic weight metric.

Data preprocessing applies demographically weighted Z-score standardization, with population size as the metric weight so that each region contributes proportionally to its demographic scale. Prior to the three-way analysis, multivariate feasibility is verified for every yearly table using the Kaiser-Meyer-Olkin (KMO $\geq$ 0.5) measure of sampling adequacy and Bartlett's test of sphericity ($p < 0.05$).

The STATIS algorithm proceeds through the following stages: (1) construction of the intra-structure (cross-product) matrix $W_k$ for each year from the standardized data; (2) inter-structure analysis using the RV coefficient (Escoufier, 1973) between all pairs of years, assembled into matrix H; (3) derivation of the optimal compromise weights αk from the first eigenvector of H; (4) synthesis of the compromise matrix $W_c$ as the α-weighted sum of the yearly matrices, followed by Singular Value Decomposition to obtain compromise scores and loadings under two weighting scenarios (the classical ACT with regions as host, and the dual STATIS with indicators as host); and (5) validation of

the compromise structure through a Monte Carlo permutation test with B = 1000 iterations on the first eigenvalue of H (Abdi et al., 2012; Phipson & Smyth, 2010), complemented by the structure-vector analysis $\tilde{\beta} = \lambda_1 \alpha_1$ under the rank-1 probabilistic model of Areia et al. (2008).

Spatio-temporal dynamics are examined by projecting each yearly table into the compromise space (Dynamic Biplot) to trace regional trajectories and annual centroid shifts. Advanced grouping applies the K-Means algorithm (MacQueen, 1967) on the compromise coordinates, with the optimal number of clusters determined jointly by the Elbow method (Syakur et al., 2018), silhouette coefficients (Rousseeuw, 1987), and Ward's agglomerative dendrogram (Ward, 1963); cluster labels are aligned across years using the Hungarian algorithm (Kuhn, 1955) and validated externally against the UNDP HDI categories. Spatial autocorrelation of the compromise scores is tested using Global Moran's I (Anselin, 1988). The entire computational pipeline is implemented in Python and integrated into an interactive web dashboard built with the Dash and Plotly frameworks (Plotly Technologies Inc., 2024). The dashboard source code is publicly available at https://huggingface.co/spaces/ammarhnfi/skripsi/blob/main/app.py.

## 3. Result and Discussion

This section presents the results of the longitudinal analysis of 41 regencies/municipalities in Banten, DKI Jakarta, and West Java Provinces for 2020–2024 using the STATIS method, covering data exploration and feasibility tests, inter-structure analysis, compromise structure construction, stability testing, spatio-temporal trajectory analysis, cluster analysis, and spatial autocorrelation testing.

### 3.1 Data Exploration and Multivariate Feasibility Tests

Descriptive exploration reveals dynamics consistent with the pandemic context. The median Open Unemployment Rate spiked in 2020 due to the COVID-19 pandemic and then declined consistently through 2024, accompanied by a narrowing distribution that indicates increasingly even labor-market improvement. The distribution of PDRB ADHK is highly right-skewed, with extreme outliers consistently representing the DKI Jakarta municipalities, while the Poverty Line rose with a widening distribution range from year to year owing to inflationary pressure. The Percentage of Poor Population declined gradually from 2020 to 2024, reflecting post-pandemic poverty alleviation, and the HDI increased consistently across all observed regions.

Before applying STATIS, multivariate feasibility was tested using the KMO measure and Bartlett's test of sphericity for each year. The results are summarized in Table 1.

Table 1. KMO and Bartlett's Test Results per Year

| Year | KMO Value | Bartlett's $\chi^2$ | df | p-value | Decision |
|---|---|---|---|---|---|
| 2020 | 0.7153 | 501.28 | 55 | < 0.001 | Adequate |
| 2021 | 0.6732 | 487.51 | 55 | < 0.001 | Adequate |
| 2022 | 0.7436 | 455.98 | 55 | < 0.001 | Adequate |
| 2023 | 0.7201 | 469.63 | 55 | < 0.001 | Adequate |
| 2024 | 0.6555 | 459.85 | 55 | < 0.001 | Adequate |

The KMO values for all years fall within the range 0.6555–0.7436 with an average of 0.7015, indicating adequate sampling for multivariate analysis. Bartlett's test yields p-values below 0.001 in every year, so the null hypothesis that the correlation matrix is an identity matrix is rejected; significant inter-indicator correlations exist and the STATIS analysis is feasible.

### 3.2 Inter-Structure Analysis

The inter-structure analysis measures the degree of similarity between the configurations of each pair of years through RV coefficients computed from the yearly intra-structure matrices. The computation results are presented in Table 2.

Table 2. RV Coefficients between Years (2020–2024)

| Year | 2020 | 2021 | 2022 | 2023 | 2024 |
|---|---|---|---|---|---|
| 2020 | 1 | 0.9760 | 0.9636 | 0.9543 | 0.9550 |
| 2021 | 0.9760 | 1 | 0.9776 | 0.9664 | 0.9626 |
| 2022 | 0.9636 | 0.9776 | 1 | 0.9772 | 0.9734 |
| 2023 | 0.9543 | 0.9664 | 0.9772 | 1 | 0.9824 |
| 2024 | 0.9550 | 0.9626 | 0.9734 | 0.9824 | 1 |

All RV coefficients exceed 0.9543, with the highest value of 0.9824 recorded for the 2023–2024 pair. This very high consistency indicates that the relative configuration of regions did not change significantly from year to year: although the COVID-19 pandemic altered the absolute values of the indicators, the hierarchy of inter-regional similarity remained intact. Relatively advanced regions remained in better positions than lagging regions, and vice versa.

The compromise weight vector α, derived from the first eigenvector of the inter-structure matrix, is distributed relatively evenly across years, ranging from 0.1938 to 0.2076 under the region-host scenario. This balance confirms that no single year dominates or distorts the compromise representation, with 2023 receiving the largest weight and therefore interpretable as the year most representative of the average 2020–2024 condition.

### 3.3 Compromise Structure Analysis

The compromise matrix was constructed through optimal spectral weighting and decomposed using Singular Value Decomposition. The first two dimensions explain 71.88% of the total variance: Dimension 1 with an eigenvalue of 6.6934 (60.85%) and Dimension 2 with an eigenvalue of 1.2132 (11.03%). This cumulative value exceeds the 70% threshold commonly used in dimension-reduction analysis, and the scree plot exhibits a clear elbow after Dimension 2, so the two-dimensional representation is considered sufficiently informative. The compromise loadings of each indicator are presented in Table 3.

Table 3. Compromise Loadings of the Indicators (Scenario 1)

| Indicator | PC1 | PC2 |
|---|---|---|
| IPG | 0.8248 | −0.2499 |
| PPM | −0.8816 | 0.0509 |
| PDRB ADHK | 0.7390 | −0.1576 |
| RLS | 0.9538 | −0.0801 |
| AHH L | 0.8891 | 0.0982 |
| AHH P | 0.8831 | 0.1055 |

| Indicator | PC1 | PC2 |
|---|---|---|
| GK | 0.8830 | −0.1500 |
| PKKP | −0.8497 | 0.1062 |
| TPT | 0.2007 | 0.8620 |
| TPAK | −0.4764 | −0.4946 |
| GR | 0.6229 | 0.1391 |

Based on the loading pattern in Table 3, Dimension 1 is interpreted as the "Level of Human and Economic Development Progress". The indicators with the strongest positive loadings are RLS (0.9538), Male Life Expectancy (0.8891), Female Life Expectancy (0.8831), GK (0.8830), IPG (0.8248), and PDRB ADHK (0.7390), whereas PPM (−0.8816) and PKKP (−0.8497) carry strong negative loadings. Regions with high PC1 scores excel in educational attainment, health, and economic productivity while exhibiting low poverty. Dimension 2 is interpreted as "Labor Market Pressure", dominated by TPT (+0.8620) with a moderate negative contribution from TPAK (−0.4946).

In the compromise plane, the DKI Jakarta administrative municipalities occupy the right side of Dimension 1 — with the highest score recorded by Kota Jakarta Selatan (4.5753) — consistent with the positive loadings of RLS, AHH, and PDRB ADHK, whereas Kabupaten Pandeglang (−5.3232) and Lebak (−5.0109) occupy the far left, aligned with the loadings of PPM and PKKP. The gradation of regional progress is thus read as a continuum along Dimension 1, as visualized in the compromise biplot (Fig. 1).

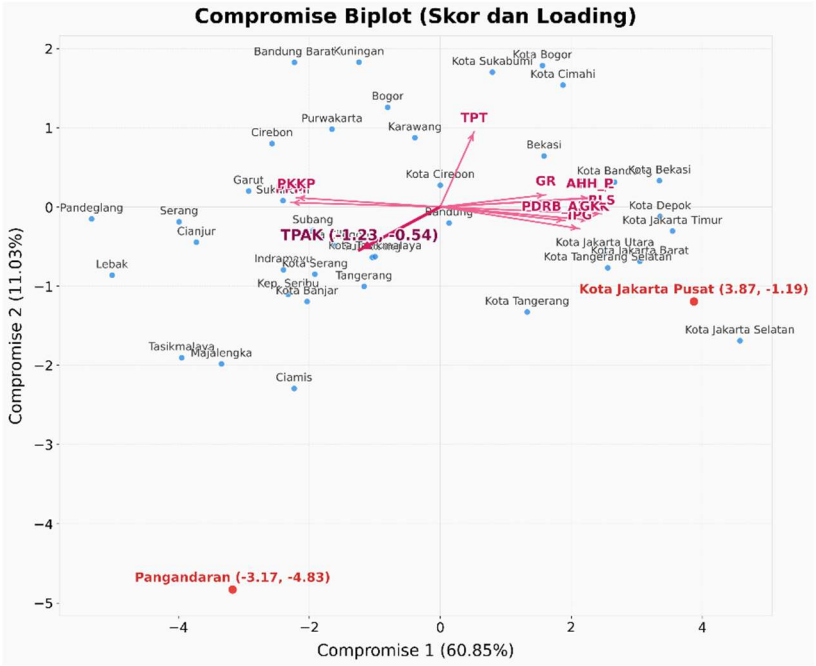



Fig. 1. Compromise biplot: structural relationships among regions and indicators (Scenario 1)

## 3.4 Stability Test of the Compromise Structure

The validity of the compromise structure was examined through a Monte Carlo permutation test with B = 1000 iterations, in which the entries of each indicator column were shuffled independently for every year to destroy inter-indicator correlations. The observed eigenvalue $\lambda_1 = 6.6934$ far exceeds the entire null distribution, with a p-value of $0.0010 < 0.05$, so the null hypothesis is rejected: the compromise structure is

statistically significant and not the product of random fluctuation (Fig. 2). The rank-1 diagnostic yields a first-dimension inter-structure inertia of 97.52%, confirming that a single dominant component suffices to summarize the inter-structure information.

The structure-vector analysis $\beta = \lambda_1 \alpha_1$ records three distinct temporal phases: (i) pandemic stagnation in 2020–2021 with nearly identical values (103.03 and 102.98), showing that the pandemic shock was uniform with respect to the relative hierarchy of regions; (ii) progressive recovery in 2022–2023 with a gradual rise to 106.84 and a peak of 110.32; and (iii) stabilization in 2024 with a slight moderation to 108.31, indicating structural consolidation rather than decline. The transversal analysis by province reveals heterogeneous temporal responses: DKI Jakarta rose consistently with the largest acceleration in 2023–2024 (+10.2%), West Java was the most stable, and Banten was the most volatile with an +18.9% surge in 2023–2024.

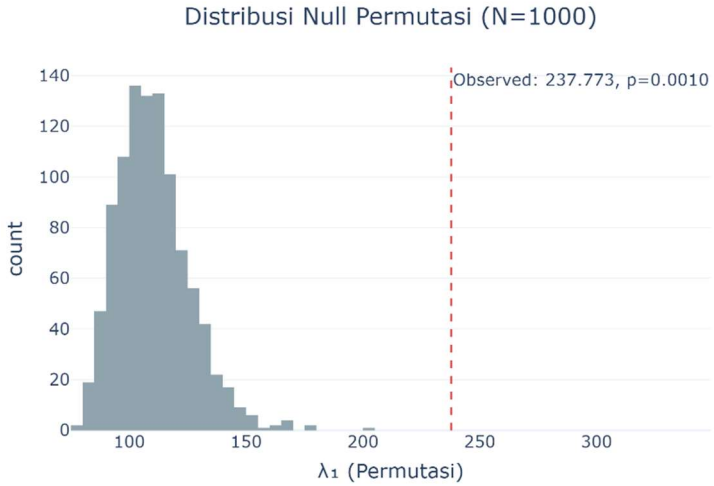



Fig. 2. Null distribution of the permutation test against the observed eigenvalue λ1obs

## 3.5 Spatio-Temporal Dynamics (Trajectory) Analysis

Trajectory analysis projects each yearly table into the fixed compromise space, so that the movement path of every region can be traced from 2020 to 2024 (Fig. 3). Three main trajectory patterns are identified: a V-shaped rebound in service-based metropolitan areas, which declined in 2020–2021 due to the pandemic contraction but returned to or exceeded their initial positions in 2022–2024; gradual monotone improvement in several buffer regions; and inertia in lagging regions, whose very short trajectories indicate that macroeconomic changes did little to alter their relative positions.

The largest positive PC1 shifts over 2020–2024 were recorded by Kota Tangerang Selatan (+0.85, from 2.23 to 3.09), Kota Bekasi (+0.39), and Kota Jakarta Pusat (+0.26), confirming strong recovery in service-based metropolitan municipalities. Conversely, the largest negative shifts occurred in Kabupaten Pangandaran (−1.37, from −2.60 to −3.97), Garut (−0.61), and Cianjur (−0.51), indicating that the inland southern regencies experienced relative decline — the development gap widened rather than narrowed. The collective dynamics are corroborated by the annual centroid shift, which peaked at 0.2657 in the 2023–2024 transition, roughly 22 times larger than the smallest shift in 2020–2021 (0.0122), while per-entity displacement identifies Kabupaten Pandeglang as the most

volatile region and PDRB ADHK, RLS, and IPG as the structural anchors of the compromise axes.

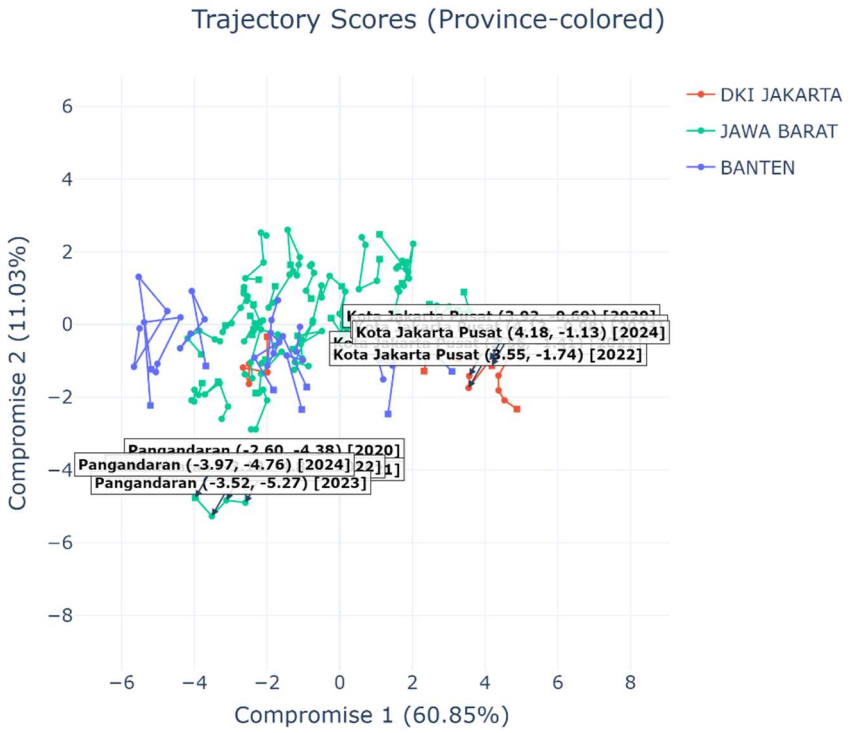



Fig. 3. Spatio-temporal trajectories of regional socio-economic positions, 2020–2024

### 3.6 Spatio-Temporal Cluster Analysis

The optimal number of clusters was determined through Elbow curves, which break consistently at k = 3 in every year of observation, as confirmed by Ward's agglomerative dendrogram. Although k = 2 yields the highest silhouette coefficients, two groups fail to distinguish the industrial buffer corridor from the lagging inland regions; k = 3, the consistent second-best solution, produces three substantively meaningful development tiers. Applying K-Means (k = 3) to the compromise scores classifies the 41 regions into the three clusters presented in Table 4 and mapped in Fig. 4.

Table 4. Cluster Membership of the STATIS Compromise (Scenario 1)

| Cluster | Number of Regions | Members (Regencies/Municipalities) |
|---|---|---|
| Cluster 1 (Underdeveloped) | 12 | Kep. Seribu, Cianjur, Garut, Tasikmalaya, Ciamis, Majalengka, Indramayu, Pangandaran, Kota Banjar, Pandeglang, Lebak, Serang |
| Cluster 2 (Developing) | 16 | Bogor, Sukabumi, Bandung, Kuningan, Cirebon, Sumedang, Subang, Purwakarta, Karawang, Bandung Barat, Kota Sukabumi, Kota Cirebon, Kota Tasikmalaya, Tangerang, Kota Cilegon, Kota Serang |
| Cluster 3 (Advanced) | 13 | Kota Jakarta Selatan, Kota Jakarta Timur, Kota Jakarta Pusat, Kota Jakarta Barat, Kota Jakarta Utara, Bekasi, Kota Bogor, Kota Bandung, Kota |

| Cluster | Number of Regions | Members (Regencies/Municipalities) |
|---|---|---|
| | | Bekasi, Kota Depok, Kota Cimahi, Kota Tangerang, Kota Tangerang Selatan |

The temporal transition matrices reveal three distinct phases of inter-cluster migration: perfect stability in 2020–2021 (zero migration, reflecting a uniform pandemic shock); mixed flows in 2021–2022 followed by selective recovery in 2022–2023, including Kota Sukabumi jumping from Cluster 2 to Cluster 3 in 2023; and full upward mobility in 2023–2024, when four regions (Kepulauan Seribu, Garut, Kota Banjar, and Kota Serang) rose from Cluster 1 to Cluster 2 without a single downward move, marking the acceleration phase of post-pandemic recovery.

External validation against the normative UNDP HDI categories shows strong structural alignment: eleven of the thirteen Cluster 3 members (84.6%) hold the "Very High" HDI classification, while most Cluster 1 members fall into the "Medium" category. All thirteen initial metropolitan members of Cluster 3 retained their position throughout 2020–2024, confirming robust structural resilience in the metropolitan core.

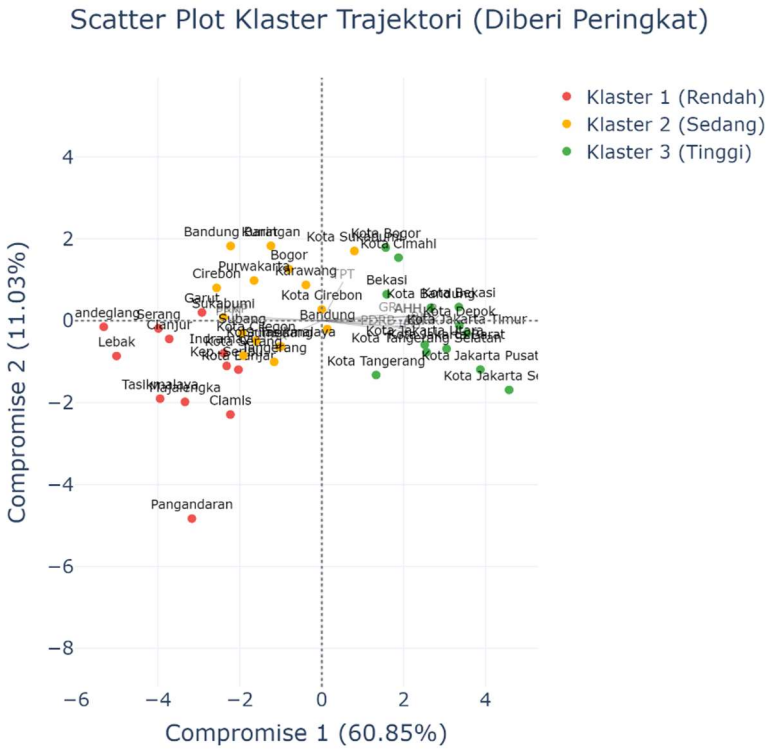



Fig. 4. Spatial distribution of the K-Means clusters (k = 3) on the compromise scores

## 3.7 Spatial Autocorrelation Test (Global Moran's I)

As a final stage, Global Moran's I was computed on the compromise PC1 scores to detect significant geographic clustering. Scenario 1 yields I = 0.6903 with a p-value of 0.3151, and Scenario 2 yields I = 0.6930 with a p-value of 0.3133; both p-values exceed 0.05, so no statistically significant spatial autocorrelation is present in the compromise scores.

The Underdeveloped–Developing–Advanced grouping is therefore a statistical cluster based on similarity of multivariate profiles, not a geographic hotspot–coldspot pattern in the classical spatial autocorrelation sense (Anselin, 1988). This finding implies that

human development performance in the Banten–DKI Jakarta–West Java corridor is determined more by internal regional capabilities and sectoral policies than by geographic proximity, so policy interventions should be targeted at the structural profiles of individual regions rather than distributed uniformly across space.

## 4. Conclusion

The STATIS method successfully constructed a stable spatio-temporal compromise representation of the 2020–2024 period, whose first two dimensions — the "Level of Human and Economic Development Progress" and "Labor Market Pressure" — explain 71.88% of the total variance. All inter-year RV coefficients above 0.9543 and a significant Monte Carlo permutation test ($p = 0.0010$) prove that the relative hierarchy of regions was persistent: the pandemic altered absolute indicator values but did not disrupt the underlying structure, and the structure vector records three phases of pandemic stagnation (2020–2021), progressive recovery (2022–2023), and stabilization (2024).

The Dynamic Biplot identifies three trajectory patterns — a V-shaped rebound in service-based metropolitan municipalities (Kota Tangerang Selatan, +0.85 on PC1), gradual progression in buffer cities such as Kota Sukabumi, and stagnation-to-relative-decline in the inland southern regencies (Kabupaten Pangandaran, −1.37) — indicating that regional development gaps widened during the observation period. K-Means clustering ($k = 3$) partitions the 41 regions into Underdeveloped (12), Developing (16), and Advanced (13) clusters that align strongly with the UNDP HDI categories, with full upward mobility in the 2023–2024 transition and complete stability of the metropolitan core.

These findings imply that uniform development policies are unlikely to be effective; targeted structural interventions are required for Cluster 1 regions — particularly Kabupaten Lebak, Pandeglang, and Garut — focusing on basic capabilities (education and health) and poverty reduction supported by regional spillover strategies from the growth core. This study is limited to eleven annual indicators in three provinces over five years; future research is advised to use higher-frequency data, extend the coverage to all of Java or Indonesia, integrate forecasting models (ARIMA, VAR, or Grey Systems GM(1,1)) on the compromise coordinate sequences, and augment the web dashboard with automatic data updates from the BPS Open Data API.

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